UNIT-1

Basic Data Mining Tasks – Data Mining Versus Knowledge Discovery in Data Bases – Data Mining Issues – Data Mining Matrices – Social Implications of Data Mining – Data Mining from Data Base Perspective.

Introduction:

Data mining is often defined as finding hidden information in a database. Alternatively, it has been called exploratory data analysis, data driven discovery, and deductive learning.

Traditional database queries (Figure 1.1), access a database using a well-defined query stated in a language such as SQL. The output of the: query consists of the data from the database that satisfies the query. The output is usually a subset of the database, but it may also be an extracted view or may contain aggregations. Data mining access of a database differs from this traditional access in several ways:

Query: The query might not be well formed or precisely stated. The data miner might not even be exactly sure of what he wants to see.

Data: The data accessed is usually a different version from that of the original operational database. The data have been cleansed and modified to better support the mining process.

Output: The output of the data mining query probably is not a subset of the database. Instead it is the output of some analysis of the contents of the database.

Data mining involves many different algorithms to accomplish different tasks. All of these algorithms attempt to fit a model to the data. The algorithms examine the data and determine a model that is closest to the characteristics of the data being examined. Data mining algorithms can be characterized as consisting of three parts:

Model: The purpose of the algorithm is to fit a model to the data.
Preference: Some criteria must be used to fit one model over another.
Search: All algorithms require some technique to search the data.

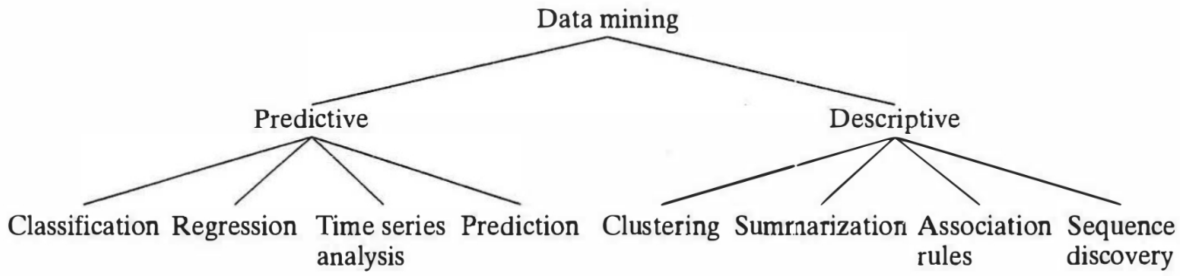

FIGURE 1.2: Data mining models and tasks.

A **predictive model** makes a prediction about values of data using known results found from different data. Predictive modeling may be made based on the use of other historical data. For example, a credit card use might be refused not because of application. the user's own credit history, but because the current purchase is similar to earlier purchases that were subsequently found to be made with stolen cards.**Example1.1**uses predictive modeling to predict the credit risk. Predictive model data mining tasks include classification, regression, time series analysis, and prediction. Prediction also be used to indicate a specific type of data mining function.

A **descriptive model** identifies patterns or relationships in data. Unlike the predictive model, a descriptive model serves as a way to explore the properties of the data examined, not to predict new properties. Clustering, summarization, association rules, and sequence discovery are usually viewed as descriptive in nature.

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1 Basic Data Mining Tasks

[*Classification, Regression, Time Series Analysis, Prediction, Clustering, Summarization, Association Rules, Sequence Discovery***]**

1.1 Classification

Classification maps data into predefined groups or classes. It is often referred to as supervised learning because the classes are determined before examining the data. Two examples of classification applications are determining whether to make a bank loan and identifying credit risks. Classification algorithms require that the classes be defined based on data attribute values. They often describe these classes by looking at the characteristics of data already known to belong to the classes. Pattern recognition is a type of classification where an input pattern is classified into one of several classes based on its similarity to these predefined classes.

EXAMPLE 1.2 An airport security screening station is used to determine: if passengers are potential terrorists or criminals. To do this, the face of each passenger is scanned and its basic pattern (distance between eyes, size and shape of mouth, shape of head, etc.) is identified This pattern is compared to entries in a database to see if it matches any patterns that are associated with known offenders

1.2 Regression

Regression is used to map a data item to a real valued prediction variable. In actuality, regression involves the learning of the function that does this mapping. Regressi?n assumes that the target data fit into some known type of functiOn (e.g., linear, logistic, etc.) and then determines the best function of this type that models the given data. some type of error analysis is used to determine which function is "best".standard hnear regression, as illustrated in Example 1.3, is a simple example of regressiOn.

EXAMPLE 1.3 A college ptofessor wishes to reach a certain level of savings before . her retirement. Periodically, she predicts what her retirement savings will be based on Its current value and several past values. She uses a simple linear regression formula to predict this. value by fitting past behavior to a linear function and then using this function to predict the values at points in the future. Based on these values, she then alters her mvestment portfolio.

1.3 Time series Analysis

With time series analysis, the value of an attribute is examined as it varies over time. The values usually are obtained as evenly spaced time points (daily, weekly, hourly, etc.). A time series plot (Figure 1.3), is used to visualize the time series. In this figure you can easily see that the plots for Y and Z have similar behavior, while X appears to have less volatility. There are three basic functions performed in time series . analysis: In one case, distance measures are used to determine the similarity between different tlme senes. In the second case, the structure of the line is examined to determine (and perhaps classify) its behavior. A third application would be to use the historical time series plot to predict future values. A time series example is given in Example 1.4.

EXAMPLE 1.4 Mr. Smith is trying to determine whether to purchase stock from Companies X, Y, or z. For a period of one month he charts the daily stock price for each company. Figure 1.3 shows the time series plot that Mr. Smith has generated. Usmg this and similar information available from his stockbroker, Mr. Sllllth decides to purchase stock X because it is less volatile while overall showing a slightly larger relative amount of growth than either of the other stocks. As a matter of fact, the stocks for Y and Z have a similar behavior. The behavior of Y between days 6 and 20 IS Identical to that for Z between days 13 and 27.

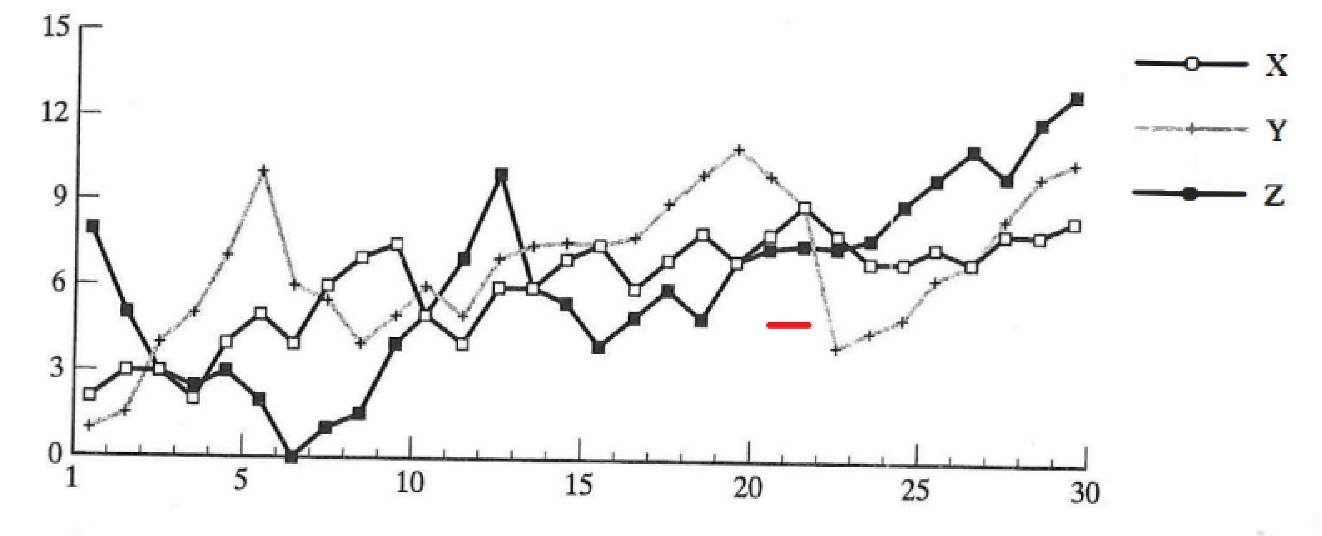

FIGURE 1.3: Time series plots.

1.4 Prediction

Many real-world data mining applications can be seen as predicting future data states based on past and current data. Prediction can be viewed as a type of classification. (Note: This is a data mining task that is different from the prediction model, although the prediction task is a type of prediction model.) The difference is that prediction is predicting a future state rather than a current state. H ere we are referring to a type of application rather than to a type of data mining modeling approach, as discussed earlier. Prediction applications include flooding, speech recognition, machine learning, and pattern recognition. Although future values may be predicted using time series analysis or regression techniques, other approaches may be used as well. Example 1.5 illustrates the process.

EXAMPLE 1.5 Predicting flooding is a difficult problem. One approach uses monitors placed at various ; points in the river. These monitors collect data relevant to flood prediction: water level, ' rain amount, time, humidity, and so on. Then the water level at a potential flooding point in the river can be predicted based on the data collected by the sensors upriver from this point. The prediction must be made with respect to the time the data were collected.

1.5 Clustering

Clustering is similar to classification except that the groups are not predefined, but rather defined by the data alone. Clustering is alternatively referred to as unsupervised learning or segmentation. It can be thought of as partitioning or segmenting the data into groups that might or might not be disjointed. The clustering is usually accomplished by determining the similarity among the data on predefined attributes. The most similar data are grouped into clusters. Example 1.6 provides a simple clustering example. Since the clusters are not predefined, a domain expert is often required to interpret the meaning of the created clusters.

EXAMPLE 1.6 A certain national department store chain creates special catalogs targeted to various demographic groups based on attributes such as income, location, and physical characteristics of potential customers (age, height, weight, etc.).To determine the target mailings of the various catalogs and to assist in the creation of new, more specific catalogs, the company performs a clustering of potential customers based on the determined attribute values. The results of the clustering exercise are then used by management to create special catalogs and distribute them to the correct target population based on the cluster for that catalog.

A special type of clustering is called segmentation. With segmentation a database is partitioned into disjointed groupings of similar tuples called segments. Segmentation is often viewed as being identical to clustering. In other circles segmentation is viewed as a specilic type of clustering applied to a database itself. In this text we use the two terms, clustering and segmentation, interchangeably.

1.6 Summarization

Summarization maps data into subsets with associated simple descriptions. Summarization is also called characterization or generalization. It extracts or derives representative information about the database. This may be accomplished by actually retrieving portions of the data. Alternatively, summary type information (such as the mean of some numeric attribute) can be derived from the data. The summarization succinctly characterizes the contents of the database. Example 1.7 illustrates this process.

EXAMPLE 1.7 One of the many criteria used to compare universities by the U.S. News & World Report is the average SAT or AC T score [GM99]. This is a summarization used to estimate the type and intellectual level of the student body.

1.7 Association Rules

Link analysis, alternatively referred to as affinity analysis or association, refers to the data mining task of uncovering relationships among data. The best example of this type of application is to determine association rules. An association rule is a model that identifies specific types of data associations. These associations are often used in the retail sales community to identify items that are frequently purchased together. Example 1.8 illustrates the use of association rules in market basket analysis. Here the data analyzed consist of information about what items a customer purchases. Associations are also used in many other applications such as predicting the failure of telecommunication switches.

EXAMPLE 1.8 A grocery store retailer is trying to decide whether to put bread on sale. To help determine the impact of this decision, the retailer generates association rules that show what otherproducts are frequently purchased with bread. He finds that 60% of the time that bread is sold so are pretzels and that 70% of the time jelly is also sold. Based on these facts, he tries to capitalize on the association between bread, pretzels, and jelly by placing some pretzels and jelly at the end of the aisle where the bread is placed. In addition, he decides not to place either of these items on sale at the same time.

1.8 Sequence Discovery

Sequential analysis or sequence discovery is used to determine sequential patterns in data. These patterns are based on a time sequence of actions. These patterns are similar to associations in that data (or events) are found to be related, but the relationship is based on time. Unlike a market basket analysis, which requires the items to be purchased at the same time, in sequence discovery the items are purchased over time in some order. Example 1.9 illustrates the discovery of some simple patterns. A similar type of discovery can be seen in the sequence within which data are purchased. For example, most people who purchase CD players may be found to purchase CDs within one week. As we will see, temporal association rules really fall into this category.

EXAMPLE 1.9 The Webmaster at the XYZ Corp. periodically analyzes the Web log data to determine how users of the XYZ's Web pages access them. He is interested in determining what sequences of pages are frequently accessed. He determines that 70 percent of the users of page A follow one of the following patterns of behavior: (A, B, C) or (A, D, B, C) or (A, E, B, C). He then determines to add a link directly from page A to page C.

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2. Data mining versus knowledge discovery in databases

The terms knowledge discovery in databases (KDD) and data mining are often used interchangeably. In fact, there have been many other names given to this process of discovering useful (hidden) patterns in data: knowledge extraction, information discovery, exploratory data analysis, information harvesting, and unsupervised pattern recognition. Over the last few years KDD has been used to refer to a process consisting of many steps, while data mining is only one of these steps.

DEFINITION Knowledge discovery in databases (KDD) is the process of finding useful information and patterns in data.

DEFINITION Data mining is the use of algorithms to extract the information and patterns derived by the KDD process.

KDD is a ' process that involves many different steps. The input to this process is the data, and the output is the useful information desired by the users. However, the objective may be unclear or inexact. The process itself is interactive and may require much elapsed time. To ensure the usefulness and accuracy of the results of the process, interaction throughout the process with both domain experts and technical experts might be needed. Figure 1.4 (modified from [FPSS96c]) illustrates the overall KDD process.

The KDD process consists of the following five steps:

- **Selection:** The data needed for the data mining process may be obtained from many different and heterogeneous data sources. This first step obtains the data from various databases, files, and nonelectronic sources.
- **Preprocessing:** The data to be used by the process may have incorrect or missing data. There may be anomalous data from multiple sources involving different data types and metrics. There may be many different activities performed at this time. Erroneous data may be corrected or removed, whereas missing data must be supplied or predicted (often using data mining tools).
- **Transformation:** Data from different sources must be converted into a common format for processing. Some data may be encoded or transformed into more usable formats. Data reduction may be used to reduce the number of possible data values being considered. • Data mining: Based on the data mining task being performed, this step applies algorithms to the transformed data to generate the desired results.
- **Interpretation/evaluation:** How the data mining results are presented to the users is extremely important because the usefulness of the results is dependent on it. Various visualization and GUI strategies are used at this last step.

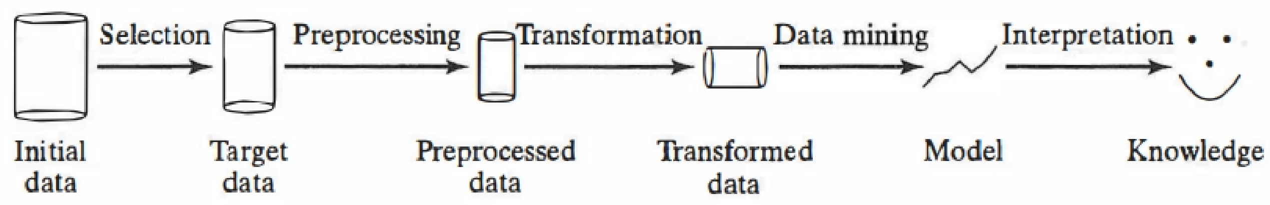

FIGURE 1.4: KDD process (modified from [FPSS96c]).

As with all steps in the KDD process, however, care must be used in performing transformation. If used incorrectly, the transformation could actually change the data such that the results of the data mining step are inaccurate.

Visualization refers to the visual presentation of data. The old expression "a picture is worth a thousand words" certainly is true when examining the structure of data. For example, a line graph that shows the distribution of a data variable is easier to understand and perhaps more informative than the formula for the corresponding distribution. The use of visualization techniques allows users to summarize, extra.ct, and grasp more complex results than more mathematical or text type descriptions of the results.

Visualization techniques include:

- **Graphical:** Traditional graph structures including bar charts, pie charts, histograms, and line graphs may be used.
- **Geometric:** Geometric techniques include the. box plot and scatter diagram techniques.
- **Icon-based:** Using figures, colors, or other icons can improve the presentation of the results.
- **Pixel-based:** With these techniques each data value is shown as a uniquely colored pixel.
- **Hierarchical:** These techniques hierarchically divide the display area (screen) into regions based on data values.
- **Hybrid:** The preceding approaches can be combined into one display.

Any of these approaches may be two-dimensional or three-dimensional. Visualization tools can be used to summarize data as a data mining technique itself. In addition, visualization can be used to show the complex results of data mining tasks

2.2. The Development of Data Mining

The current evolution of data mining functions and products is the result of years of influence from many disciplines, including databases, information retrieval, statistics, algorithms, and machine learning (Figure 1 .5). Another computer science area that has had a major impact on the KDD process is multimedia and graphics. A major goal of KDD is to be able to describe the results of the KDD process in a meaningful manner. Because many different results are often produced, this is a nontrivial problem. Visualization techniques often involve sophisticated multimedia and graphics presentations. In addition, data mining techniques can be applied to multimedia applications.

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3. Data mining issues

There are many important implementation issues associated with data mining:

1. **Human interaction:** Since data mining problems are often not precisely stated, interfaces may be needed with both domain and technical experts. Technical experts are used to formulate the queries and assist in interpreting the results. Users are needed to identify training data and desired results.
2. **Overfitting:** When a model is generated that is associated with a given database state it is desirable that the model also fit future database states. Overfitting occurs where the model does not fit future states. This may be caused by assumptions that are made about the data or may simply be caused by the small size of the training database.Overfitting can arise under other circumstances as well, even though the data are not changing.
3. **Outliers:** There are often many data entries that do not fit nicely into the derived model. This becomes even more of an issue with very large databases. If a model is developed that includes these outliers, then the model may not behave well for data that are not outliers.
4. **Interpretation of results :** Currently, data mining output may require experts to correctly interpret the results, which might otherwise be meaningless to the average database user.

5. **Visualization of results:** To easily view and understand the output of data mining algorithms, visualization of the results is helpful.
6. **Large datasets:** The massive datasets associated with data mining create problems when applying algorithms designed for small datasets. Many modeling applications grow exponentially on the dataset size and thus are too inefficient for larger datasets. Sampling and parallelization are effective tools to attack this scalability problem.
7. **High dimensionality:** A conventional database schema may be composed of many different attributes. The problem here is that not all attributes may be needed to solve a given data mining problem. In fact, the use of some attributes may interfere with the correct completion of a data mining task. The use of other attributes may simply increase the overall complexity and decrease the efficiency of an algorithm.
8. **Multimedia data:** Most previous data mining algorithms are targeted to traditional data types (numeric, character, text, etc.). The use of multimedia data such as is found in GIS databases complicates or invalidates many proposed algorithms.
9. **Missing data:** During the preprocessing phase of KDD, missing data may be replaced with estimates. This and other approaches to handling missing data can lead to invalid results in the data mining step.
10. **Irrelevant data:** Some attributes in the database might not be of interest to the data mining task being developed.
11. **Noisy data:** Some attribute values might be invalid or incorrect. These values are often corrected before running data mining applications.
12. **Changing data:** Databases cannot be assumed to be static. However, most data mining algorithms do assume a static database. This requires that the algorithm be completely rerun anytime the database changes.
13. **Integration:** The KDD process is not currently integrated into normal data processing activities. KDD requests may be treated as special, unusual, or one-time needs. This makes them inefficient, ineffective, and not general enough to be used on an ongoing basis. Integration of data mining functions into traditional DBMS systems is certainly a desirable goal.
14. **Application:** Determining the intended use for the information obtained from the data mining function is a challenge. Indeed, how business executives can effectively use the output is sometimes considered the more difficult part, not the running of the algorithms themselves. Because the data are of a type that has not previously been known, business practices may have to be modified to determine how to effectively use the information uncovered.

These issues should be addressed by data mining algorithms and products.

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4. Data mining metrics

Different metrics could be used for different techniques and also based on the interest level. From an overall business or usefulness perspective, a measure such as return on investment (ROI) could be used. ROI examines the difference between what the data mining technique costs and what the savings or benefits from its use are. Of course, this would be difficult to measure because the return is hard to quantify. It could be measured as increased sales, reduced advertising expetd . iture, or both. In a specific advertising campaign implemented via targeted catalog mailing, the percentage of catalog recipients and the amount of . purchase per rectptent would provtde one means to measure the effectiveness of the mailings.

we use a more computer science/database perspective to measure various data mining approaches. The metrics used include the traditional metrics of space. And time based on complexity analysis. In some cases, such as accuracy in classification, more specific metrics targeted to a data mining task may be used.

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5. Social implications of data mining

- The integration of data mining techniques into normal day-to-day activities has become commonplace. We are confronted daily with targeted advertising, and business has become more efficient through the use of data mining activities to reduce costs. Data mining adversaries, however, are concerned that this information is being obtained at the cost of reduced privacy.
- Data mining applications can derive much demographic information concerning customers that was previously not known or hidden in the data. The unauthorized use of such data could result in the disclosure of mformat10n that is deemed to be confidential.
- Indeed, many classification techniques work by identifying the attribute values that commonly occur for the target class. Subsequent records will be then classified based on these attribute values. Keep in mind that these approaches to classification are imperfect.
- Users of data mining techniques must be sensitive to these issues and must not violate any privacy directives or guidelines.

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6. Data mining from a database perspective

Data mining can be studied from many different perspectives. An IR researcher probably would concentrate on the use of data mining techniques to access text data; a statistician might look primarily at the historical techniques, including time series analysis, hypothesis testing, and applications of Bayes theorem; a machine learning specialist might be interested primarily in data mining algorithms that learn; and an algorithms researcher would be interested in studying and comparing algorithms based on type and complexity.

The study of data mining from a database perspective involves looking at all types of data mining applications and techniques. While our interest is not limited to any particular type of algorithm or approach,

we are concerned about the following implementation issues:

• **Scalability:** Algorithms that do not scale up to perform well with massive real world datasets are of limited application. Related to this is the fact that techniques should work regardless of the amount of available main memory.

• **Real-world data:** Real-world data are noisy and have many missing attribute values. Algorithms should be able to work even in the presence of these problems.

- **Update:** Many data mining algorithms work with static datasets. This is not a realistic assumption.
- **Ease of use:** Although some algorithms may work well, they may not be well received by users if they are difficult to use or understand.

One crucial part of the database abstraction is query processing support. One reason relational databases are so popular today is the development of SQL. It is easy to use (at least when compared with earlier query languages such as the DBTG or IMS DML) and has become a standard query language implemented by all major DBMS vendors. SQL also has well-defined optimization strategies.· Although there currently is no corresponding data mining language, there is ongoing work in the area of extending SQL to support data mining tasks.